SWE 307 BIG DATA PROJECT - 2

Redis for RDB caching and HDFS as Data Storage and a Local LLM model for chatting

Due date: 22.10.2026 Thursday, in class.

This project has multiple objectives:

1) Build a real-time information system that combines the speed of in-memory data processing with the reliability and scalability of distributed storage (See Figure 1). The system uses **Redis** to handle fast, real-time ingestion and temporary caching of incoming data.

2) Use of a distributed file server practically unlimited size in an information system as a document server (or object storage like S3 in AWS). Hadoop-HDFS is an open source distributed, fault-tolerant file system that is used for storing documents in information systems. The Hadoop HDFSis expected to improve response time as well.

3) Experimenting a local LLM server. Ollama server is added to the system just for chatting from front-end.

Key Objectives:

- Use the topic of your DIMS project from previous year for RDBMS (MySQL or PostgreSQL).
- Add **Redis** for improving reponse time as example done in the class. Show that the caching mechanism works as expected. (See example: https://www.youtube.com/watch?v=0a-RlJx09rg)
- Use **Hadoop-HDFS** for document storage, (e.g. image files). Show that images or files in the browser are coming from HDFS, or when a file is uploaded, it goes to HDFS. Please see examples done in the class.
- Install Ollama server and llama 3.2 LLM model as local LLM server. One LLM model (llama3.2:latest) would be sufficient. This server can be tested via CLI but it should communicate with your Java web application. Show that it works on the browser with a chat view like in chatGPT.
- **PS:** Installing Ollama server and llama3.2 model takes some time depending on network speed and takes a lot of disk space.

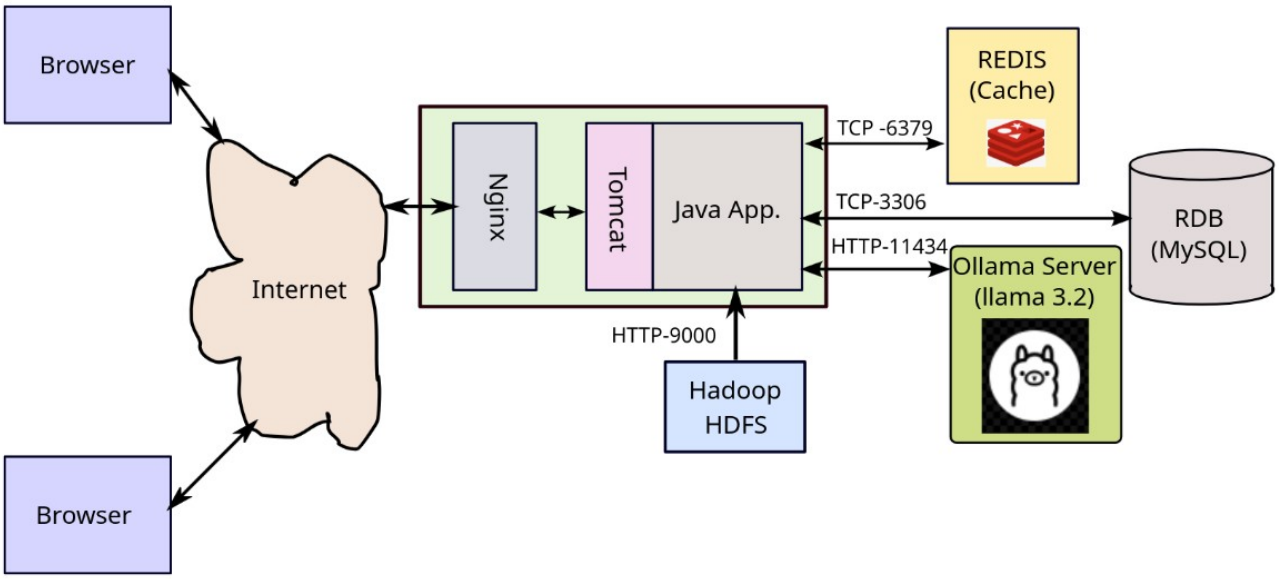


Figure 1. Architectural block diagram of project 2.

What is required from you is as follows:

1) Single node Hadoop-HDFS cluster must be installed in your computer.

2) Redis and RDBM database (MySQL or PostgreSQL) must be installed on your computer.

3) Ollama server with llama 3.2 model must be installed on your computer.

4) A simple Java Spring-Boot application will be developed to perform the following tasks:

a) Show that CRUD operations on RDBM data will be performed as expected either on database or Redis. Explain when, which and why?

b) Your application images (or files) will be stored to/read from HDFS.

c) There will be two pages in this application: 1) Application information will be displayed in a web page obtained from using some JOIN operations on database. For example information to display scott application was: employee image, employee name, manager name (self join on emp table), salary, commission (came from emp table), department name (came from dept table with join operation). Do similar work on your problem.

d) On the second page (chat page), we should be able to chat with Ollama-LLM server. Show us some chat outcome that comes from local LLM model.

Notes:

1) Some groups may want to implement the project using G-Drive or AWS-S3 Object storage instead of HDFS, these implementations will also be accepted. This will be discussed in the class.